RAJALAKSHMI ENGINEERING COLLEGE

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Y2

Started on	Friday, 25 September 2026, 8:55 AM
State	Finished
Completed on	Friday, 25 September 2026, 9:06 AM
Time taken	11 mins 17 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int ubound(int arr[], int size, int key) {
    int low = 0, high = size;
    while (low < high) {
        int mid = low + (high - low) / 2;
        if (arr[mid] <= key) {
            low = mid + 1;
        } else {
            high = mid;
        }
    }
    return low;
}

int length(int nums[], int numsSize) {
    if (numsSize == 0) return 0;


    int sub[numsSize];
    int subSize = 0;

    for (int i = 0; i < numsSize; i++) {
        int idx = ubound(sub, subSize, nums[i]);

        if (idx < subSize) {
            sub[idx] = nums[i];
        } else {
            sub[subSize++] = nums[i];
        }
    }

    return subSize;
}

int main() {
    int sequence[] = {-1, 3, 4, 5, 2, 2, 2, 2, 3};
    int n = sizeof(sequence) / sizeof(sequence[0]);

    int result = length(sequence, n);
    printf("%d\n", result);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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